

Database Management Systems

BRANCH: COMPUTER SCIENCE (CSE)

COURSE CODE: CS303

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Database Management Systems (CS303) under the Computer Science (CSE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Database System Concepts

Database vs File systems. Data abstraction, data independence. ER Model: Entities, Attributes, Relationship sets, ER diagrams. Relational Model: Relational algebra operations (Select, Project, Join, Division).

Unit 2: SQL & Query Processing

DDL, DML, DCL commands. Nested queries, Joins, Triggers, Views. Query processing stages: Parsing, Translation, Optimization, and Evaluation.

Unit 3: Normalization

Functional dependencies, closure of attributes. Normal forms: 1NF, 2NF, 3NF, BCNF. Multi-valued dependencies and 4NF. Lossless join decomposition and dependency preservation.

Unit 4: Transaction & Concurrency

Transaction states, ACID properties. Serializability: Conflict and View serializability. Concurrency control: Two-Phase Locking (2PL), Timestamp-based ordering, Deadlock handling.

ACID: Atomicity, Consistency, Isolation, Durability. 3NF: $X \rightarrow A$ implies X is superkey or A is prime.

CRITICAL FORMULA / PRINCIPLE: Key Formulas & Exam Pointers



